

Priyanshu M Sharma

+1 (480) 577-6362 | mrityunjay2627@gmail.com | linkedin.com/in/priyanshu-m-sharma | github.com/mrityunjay2627

EDUCATION

Arizona State University

Tempe, AZ

M.S. Robotics and Autonomous Systems (AI), GPA: 3.89/4.00

Aug 2024 – May 2026

- **Relevant Coursework:** Statistical Machine Learning, Deep Neural Networks, Reinforcement Learning, Artificial Intelligence for Robotics, Robotic Systems, Data Processing at Scale, Applied Linear Algebra, Bayesian Statistics

Vellore Institute of Technology

Vellore, India

B.Tech. Computer Science and Engineering and Business Systems, GPA: 3.60/4.00

July 2019 – July 2023

TECHNICAL SKILLS

Languages: Python, SQL, PySpark, C++, Java, Scala, JavaScript, R, MATLAB, ROS2, Bash

AI/ML: LangChain, RAG, MCP, LoRA, Vector Databases, PyTorch, LangSmith, Prompt Engineering, A/B Testing

Data & Analytics: Pandas, NumPy, Spark, Kafka, Airflow, dbt, Snowflake, Databricks, Neo4j, MongoDB, Power BI

Cloud & Infrastructure: Docker, Kubernetes, AWS (EC2, EBS, ECS, Fargate, EKS, S3, Glue, Kinesis, Athena, Secrets Manager), Helm, Terraform, GitHub Actions, FastAPI, Django

EXPERIENCE

Graduate Service Assistant – Collective Design (CoDe) Lab

Jan 2026 – Present

Arizona State University

Tempe, AZ

- Built **MCP-based agentic interface** to **TAT-C** aerospace toolkit, replacing manual analyst workflow with **natural-language queries**; reduced typical mission analysis from **hours to under 5 minutes** across 42 test cases.
- Conducted systematic **failure-mode analysis** on 42-query baseline (33.3% correctness): identified **coordinate hallucination** (72%), **tool ambiguity** (48%), and invalid satellite-instrument pairings (24%) as primary failure drivers; ran **ablation study** showing **ICL contributed +58pp and RAG +8pp**.
- Built **RAG-grounded location resolver** with **fuzzy matching** and confidence-scored **HITL (Human-in-the-Loop) gate** (70% threshold); deployed on **Kubernetes** with **Prometheus and Grafana** monitoring, achieving **100% correctness** across 42 queries and 67% auto-approval rate.
- Restructured 20+ low-level **Python API calls** into 4 **semantic tool wrappers**, raising end-to-end planning success from **23% to 77%** while preserving parametric flexibility for complex constellation studies.
- Ran **A/B testing on confidence thresholds** and **tool granularity strategies** to validate design choices; containerized with **Docker** and shipped on **Kubernetes** with health checks and rolling updates for production-grade availability.

Graduate Service Assistant – Yu Research Group / AMASS Lab

Sep 2025 – Dec 2025

Arizona State University

Tempe, AZ

- Teaching support for **RAS 501 (Applied Linear Algebra for Engineers)** under Prof. Xi Yu: held office hours, fielded technical questions on linear algebra and engineering applications, and graded technical assignments and examinations for accurate, consistent assessment.
- Streamlined administrative workflows for the course (assignment tracking, grade reconciliation, exam proctoring) and served as the primary point of contact for student technical inquiries.
- Contributed to AMASS Lab research on **autonomous blimp control systems** using **Python** and **ROS2**, supporting the lab's work on lighter-than-air autonomous platforms.

Consulting Analyst – Strategy & Analytics

Sep 2023 – July 2024

Deloitte USI

Bengaluru, IN

- Delivered **AI-driven contract redlining POC** for a **Fortune 50** client (*global life sciences company*); the POC outcome determined the firm's bid on the full production engagement, requiring high-quality delivery on a compressed timeline.
- Built **end-to-end RAG pipeline** over **1K+ manufacturing contracts** with mixed structured, tabular, and unstructured content: engineered **custom parsers** to handle complex multi-format extraction; benchmarked **Pinecone** and **Qdrant** for vector storage; tuned **chunking and retrieval** to preserve clause-level context.
- Designed the **redlining detection and counter-argument generation** logic alongside senior engineers; system automatically resolved **~80% of contract redlines** proposed by manufacturer counterparties, leaving only 20% edge cases for the client's legal team to review (vs. 100% manual review previously).
- Built **natural-language-to-SQL** system for an **MVP inventory and supply-chain analytics product** on the same client's data warehouse; large multi-table schemas required complex **multi-join queries** that I hand-wrote and validated during **EDA**, then encoded as **few-shot patterns and prompt templates** so the LLM could generate correct SQL from business-user queries.

- Extended the SQL system to handle **full-table extraction requests** from natural language (e.g., “give me Q3 inventory across distributors”), reducing manual analyst query time by **60%** across the supported use cases.
- Operated within **agile delivery framework**: **Jira** tickets, weekly sprints, code reviews from senior engineers; presented technical tradeoffs (latency, accuracy, cost) to senior client stakeholders across US, UK, and India.

Software Engineer Intern

June 2023 – Aug 2023

Eigengram

Gurugram, IN

- Worked alongside two technical AI founders at an early-stage **medical AI startup**, contributing to **data pipelines** and **ML model deployment** for the company’s clinical applications.
- Built **Python ETL pipeline** to parse, clean, and map **10K+ heterogeneous gene expression records** from inconsistent medical data formats; cut manual processing time by **80%** and improved downstream data quality by 25%, enabling reliable model training on previously unusable data.
- Deployed **CNN-based medical imaging models** and a **BERT-powered clinical QnA chatbot** through **Django REST APIs**; provisioned and managed **AWS EC2 and EBS** infrastructure with latency monitoring for the inference services.

PROJECTS

Distributed Transportation Analytics Pipeline | *Kubernetes, Kafka, Neo4j, Docker, Helm, Python* Dec 2025

- Built **real-time graph processing pipeline** ingesting **1,530 NYC taxi trip events** across 42 transportation zones via **Kafka** into a **Neo4j** graph database, modeling the city as a directed network of pickup/dropoff hubs.
- Ran **PageRank** to identify high-traffic transit hubs and **BFS** for shortest-path routing between zones using the **Neo4j Graph Data Science** library; identical analytical results across both monolithic **Docker** and distributed **Kubernetes** deployments validated correctness.
- Migrated from single-container Docker to **Kubernetes microservices** (Zookeeper, Kafka, Neo4j, Sink Connector); resolved a hardcoded-credential bug in the pre-built connector image via **ConfigMap injection**, raising deployment success from **90% to 100%**. Architecture transferable to **logistics, fleet routing, and supply-chain network analysis**.

Autonomous Hierarchical Multi-Agent System | *LangChain, LangSmith, Docker, ECS, GitHub Actions* Aug 2025

- Designed **orchestrator-worker agent architecture** in **LangChain** to address single-agent complexity and hallucination: a planner decomposes intent and worker agents execute tools, enabling separation of concerns and multi-step debugging.
- Shipped as a persistent, self-healing service on **AWS ECS with Fargate** – containerized via **Docker** and automated through **GitHub Actions CI/CD**, eliminating manual image-push and restart cycles between iterations.
- Implemented **security-first credentials** with **AWS Secrets Manager** and **least-privilege IAM** (no API keys in code or env files); instrumented with **LangSmith** for trace-level debugging of agent reasoning.

Sentinel-Robot – AMR Fleet DataOps Pipeline | *SQL, dbt, AWS, Terraform, GitHub Actions* Aug 2025

- Built **DataOps pipeline** transforming raw AMR telemetry (motor temperature, vibration, battery, position) into aggregated health reports, replacing reactive break-fix cycles with a validated **single source of truth** for **warehouse fleet operators**.
- Designed the full stack as **IaC: Kinesis Firehose** to S3 data lake, **dbt** staging-to-marts with **star-schema fact tables**, and **Amazon Managed Workflows for Apache Airflow**-orchestrated DAGs chaining **Glue Crawlers** and dbt runs; **Terraform** provisioning and **GitHub Actions CI/CD** cut code-to-production from 10-15 min manual to ~2 min automated and reduced operator intervention by **~95%**.
- Wrote **advanced SQL** across the stack: **window functions** and aggregations in **dbt models**, generic and custom **dbt tests** for schema and business-threshold validation, and partitioned **Athena** queries on pre-aggregated **Parquet** marts – cut query latency from 3.2s on raw JSON to 0.7s (**~78% reduction**) for downstream BI consumption.

LLaMA 2 Optimization with LoRA and RAG | *PyTorch, LangChain, LoRA, SentenceTransformers* Apr 2025

- Implemented **LLaMA 2 (7B) from scratch** in **PyTorch** with **Grouped Multi-Query Attention**, **KV caching**, **RoPE**, and **SwiGLU**; achieved **20% inference speedup** on CPU vs. baseline.
- Added **LoRA adapters (rank 4)** freezing 99% of weights for parameter-efficient fine-tuning; reached **18.68 BLEU** on Czech-English translation, beating OPT-350M and GPT-7B baselines.
- Built **RAG layer** with **sentence-transformer embeddings** and cosine retrieval over chunked document corpora to ground answers and reduce hallucination on technical Q&A.

Digital Twin – 6-DOF Collaborative Robot | *MATLAB Simscape, Python, OpenCV, ArUco, TCP/IP* Dec 2024

- Owned the **computer vision** and **MATLAB digital-twin workstream** on a multi-disciplinary team (with electrical and mechanical engineers) for the MyCobot Pro 600 industrial arm.
- Wrote the **MATLAB control and kinematics code**, including **DH-parameter forward kinematics** – for **Simscape multibody digital twin**; validated planned trajectories against joint limits and workspace before sending to hardware.
- Implemented **OpenCV ArUco marker detection** for vision-guided motion planning; verified sim-to-real end-effector positioning over a Python **TCP/IP control link** to the robot’s onboard OS.